



Beam centerline / projected section

$$\delta W = \left(\mathbf{f}_d^{global} \right)^T \delta \mathbf{u}_d^{global} = \left(\mathbf{Q}_o^{global} \right)^T \delta \mathbf{q}_o^{global},$$

$$\mathbf{Q}_o^{global} = \mathbf{H}_{ds}^T \mathbf{f}_d^{global}$$